Answer Key & Teacher Notes

Session 1 — Area of Triangles · Session 2 — Missing Measures in Triangles

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Area of Triangles

TALK PROMPTS

- Why does the triangle formula include a half? Explain with two copies of the triangle.
- Why does halving the product give the triangle and not the parallelogram?

PARTNER TASK

Cut Sandwich and Right-Triangle Sail. One partner cuts a paper rectangle on a diagonal and shows the two congruent triangles; the other writes $A = \frac{1}{2} \times b \times h$ and solves. Switch roles with Recess Wedge: base 14 ft, height 9 ft (63 ft²).

WATCH FOR

- Forgetting to take half (finding the parallelogram's area).
- Using a slanted side as the height.
- Not seeing that the legs of a right triangle are the base and height.

SESSION 2 · TEACHER NOTES

Missing Measures in Triangles

TALK PROMPTS

- How is finding a missing base different from finding an area? What changes?
- What would change in your steps if the height were given instead of the base?

PARTNER TASK

Quilt Piece and Roof Gable. One partner substitutes into $A = \frac{1}{2} \times b \times h$; the other doubles the area, divides, and checks by working forward. Switch roles with Awning Panel.

WATCH FOR

- Dividing the area by one measure without doubling ($66 \div 11 = 6$).
- Writing a length answer in square units (12 ft²).
- Halving the area instead of doubling it to undo the ½.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.G.A.1	1	2	1	1	1	2	2	3	2	1	1
S1 · Form B	6.G.A.1	1	2	1	1	1	2	2	3	2	1	1
S2 · Form A	6.G.A.1	1	2	1	1	1	2	2	3	2	1	1
S2 · Form B	6.G.A.1	1	2	1	1	1	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

right triangle: A triangle with one **square** corner.

height: The distance from the base to the opposite corner, at a **right** angle.

congruent: Exactly the same **size** and shape.

- Two congruent triangles make a **parallelogram**.
- So a triangle's area is **half** of base × height: A = ½ × b × h.
- In a right triangle, the two **legs** are the base and the height.

Example: 14 × 9 = **126** 126 cm² is the area of the parallelogram made by two triangles. 126 ÷ 2 = **63** Answer: The area is **63** cm².

YOU TRY

1. a. 35 m² b. 20 ft²

2. ½ × 12 × 5 = 30 cm² (the 13 cm side is not used)

EXIT TICKET

1. a. 99 in² b. 14 cm²

2. ½ × 9 × 6 = 27 in²

3. **B** (A = ½ × b × h)

PRACTICE FORM A

1. a. 30 cm² b. 21 in² c. 70 ft² d. 44 m²

2. a. 12 × 20 = 240 in² b. 240 ÷ 2 = 120 in²

3. a. ½ × 10 × 8 = 40 cm² b. 2 × 40 = 80 cm²

4. ½ × 6 × 11 = 33 ft²

5. ½ × 30 × 26 = 390 cm²

6. ½ × 14 × 9 = 63 ft²

7. P: ½ × 18 × 10 = 90 m². Q: ½ × 12 × 13 = 78 m². Banner P covers more.

8. a. ½ × 24 × 15 = 180 ft² b. 180 ÷ 30 = 6 bags c. 6 × $6 = $36. If the base doubles, the area doubles to 360 ft², so the cost doubles to $72.

9. 220 cm² is the area of the parallelogram made by two triangles. He forgot to take half: A = ½ × 22 × 10 = 110 cm².

10. **B** (42 in²)

11. **C** (1/2)

Self-check word **HALVING**

PRACTICE FORM B

1. a. 40 cm² b. 18 in² c. 66 ft² d. 52 m²

2. a. 10 × 18 = 180 in² b. 180 ÷ 2 = 90 in²

3. a. ½ × 12 × 9 = 54 cm² b. 2 × 54 = 108 cm²

4. ½ × 8 × 15 = 60 ft²

5. ½ × 24 × 21 = 252 cm²

6. ½ × 16 × 7 = 56 ft²

7. P: ½ × 16 × 9 = 72 m². Q: ½ × 12 × 14 = 84 m². Banner Q covers more.

8. a. ½ × 20 × 18 = 180 ft² b. 180 ÷ 30 = 6 bags c. 6 × $5 = $30. If the base doubles, the area doubles to 360 ft², so the cost doubles to $60.

9. 140 m² is the area of the parallelogram made by two triangles. She forgot to take half: A = ½ × 20 × 7 = 70 m².

10. **B** (40 m²)

11. **A** (a square corner (right angle))

Self-check word **PRODUCT**

GUIDED NOTES

area: The space inside a shape, measured in **square** units.

formula: A rule written with **letters** and symbols.

inverse operation: An operation that **undoes** another.

- Undo the half first: **double** the area.
- Then **divide** by the measure you know.
- A base or height is a length, so use units like cm, not **cm²**.

Example: 40 = ½ × 10 × *h* Double the area: 2 × 40 = **80** Divide by the base: 80 ÷ 10 = **8** Answer: The height is **8** cm.

YOU TRY

1. a. 10 ft b. 6 m

2. 2 × 48 = 96; 96 ÷ 16 = 6 m

EXIT TICKET

1. a. 12 cm b. 18 in

2. 2 × 35 = 70; 70 ÷ 10 = 7 ft

3. **B** (half as big as it should be)

PRACTICE FORM A

1. a. 9 cm b. 8 in c. 10 m d. 9 ft

2. a. 126 = ½ × b × 18 b. 2 × 126 = 252; 252 ÷ 18 = 14 yd

3. 2 × 54 = 108; 108 ÷ 12 = 9 in.

4. 2 × 117 = 234; 234 ÷ 9 = 26 ft

5. 2 × 60 = 120; 120 ÷ 15 = 8 ft

6. No. ½ × 28 × 7 = 98, not 84. Base = 2 × 84 ÷ 7 = 168 ÷ 7 = 24 cm.

7. 2 × 45 = 90; 90 ÷ 5 = 18 m

8. a. 2 × 120 ÷ 16 = 15 ft b. 2 × 120 ÷ 12 = 20 ft c. Many base–height pairs multiply to 240 (16 × 15 and 20 × 12), so the triangles have the same area but different shapes.

9. He skipped undoing the ½. Double the area first: 2 × 66 = 132, and 132 ÷ 11 = 12 ft. Check: ½ × 11 × 12 = 66.

10. **B** (12 cm)

11. **C** (10 in.)

Self-check word **MISSING**

PRACTICE FORM B

1. a. 8 cm b. 8 in c. 10 m d. 6 ft

2. a. 132 = ½ × b × 12 b. 2 × 132 = 264; 264 ÷ 12 = 22 yd

3. 2 × 42 = 84; 84 ÷ 7 = 12 in.

4. 2 × 135 = 270; 270 ÷ 15 = 18 ft

5. 2 × 70 = 140; 140 ÷ 14 = 10 ft

6. No. ½ × 10 × 9 = 45, not 90. Base = 2 × 90 ÷ 9 = 180 ÷ 9 = 20 ft.

7. 2 × 52 = 104; 104 ÷ 8 = 13 m

8. a. 2 × 150 ÷ 20 = 15 ft b. 2 × 150 ÷ 10 = 30 ft c. Many base–height pairs multiply to 300 (20 × 15 and 30 × 10), so the triangles have the same area but different shapes.

9. He skipped undoing the ½. Double the area first: 2 × 90 = 180, and 180 ÷ 15 = 12 cm. Check: ½ × 15 × 12 = 90.

10. **B** (8 m)

11. **C** (10 in.)

Self-check word **INVERSE**

SHEET 1 · AREA OF TRIANGLES

1. ½ × 12 × 7 = 42 cm²
2. ½ × 9 × 8 = 36 in²
3. a. ½ × 10 × 9 = 45 cm² b. ½ × 5 × 11 = 27.5 ft²
4. Area = ½ × 15 × 6 = 45 ft²; 45 ÷ 9 = 5 bags
5. He forgot to take half (40 m² is the parallelogram's area). Correct area: ½ × 8 × 5 = 20 m².

SHEET 2 · MISSING MEASURES IN TRIANGLES

1. A: h = 2 × 28 ÷ 8 = 7 in; B: b = 2 × 54 ÷ 9 = 12 ft
2. 2 × 48 = 96; 96 ÷ 16 = 6 m
3. a. 2 × 40 ÷ 10 = 8 m b. 2 × 27 ÷ 6 = 9 in
4. 2 × 35 = 70; 70 ÷ 10 = 7 ft
5. Neither. P = ½ × 12 × 5 = 30 cm² and Q = ½ × 6 × 10 = 30 cm², so the areas are equal.
6. **C** (12 units)
7. **B** (½ × 18 × 5)

October 2026 conceptual additions — teacher key

Session 1 · Area of Triangles

Parallelogram area 60 cm²; one triangle 30 cm². Each triangle is half of the parallelogram.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Parallelogram area 60 cm²; one triangle 30 cm². Each triangle is half of the parallelogram.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Missing Measures in Triangles

36 = 1/2 × 9 × h; h = 72/9 = 8 cm. Check 1/2 × 9 × 8 = 36 cm².

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: 36 = 1/2 × 9 × h; h = 72/9 = 8 cm. Check 1/2 × 9 × 8 = 36 cm².

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.